Thermistor String Buoy on Meereisportal.de

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Photo: AWI

Description of autonomous platform

Thermistor string buoys (also referred to as ice mass balance buoys, IMBs) measure temperature and thermal conductivity through air, snow, sea ice and seawater. They are also often called SAMS IMBs, referring to the developer (Scottish Association for Marine Science). The 4.8 m long thermistor string features 240 embedded thermistors at a spacing of 2 cm (custom length/spacing also available) and 3 m tail to aid deployment. The temperature measurements and the response to cyclic heating of resistors enable the determination of snow depth and sea-ice thickness. The data are transmitted via Iridium at varying intervals.

Property name	Property value
Manufacturer	SRSL, Oban, UK
Data provider	SIMBA (SAMS Ice Mass Balance buoy)
Weight	approx. 20 kg (depends on battery package)
Deployment type	Installation on sea ice, chain lowered through 5 cm auger hole
Power supply	Alkaline batteries (life time >1 year)
Data transmission interval	2 hour (user defined)

Parameters and accuracy

Parameter	Sensor	Accuracy
Temperature (˚C)	240 x Maxim DS28EA00	calibrated to: ± 0.125°C at 1°C
Barometric pressure (hPa)	Freescale Semiconductor MPL115A1	± 1 hPa
Position (from GPS)	Fastrax UP501 GPS receiver	± 2 m

Data Processing

The data set has been processed and contains quality flags for different kinds for erroneous data. Flag values are the sum of individual error codes. The value of 0 refers to no error.

Quality flag, position: The geographic position is flagged +1 if the drift velocity, as derived from the GPS longitude and latitude, exceeds a threshold of 10 deg latitude or 50 deg longitude per time step; +2 if the position exceeds extreme values, such as longitude > 360 deg; +4 if the position is exactly 0.0.

How to cite

1. The reference for the (data portal of) Meereisportal.de is:

Grosfeld, K.; Treffeisen, R.; Asseng, J.; Bartsch, A.; Bräuer, B.; Fritzsch, B.; Gerdes, R.; Hendricks, S.; Hiller, W.; Heygster, G.; Krumpen, T.; Lemke, P.; Melsheimer, C.; Nicolaus, M.; Ricker, R. and Weigelt, M. (2016), Online sea-ice knowledge and data platform <www.meereisportal.de>, Polarforschung, Bremerhaven, Alfred Wegener Institute for Polar and Marine Research & German Society of Polar Research, 85 (2), 143-155, doi:10.2312/polfor.2016.011.

2. Please include the following phrase into the acknowledgment, substituting PARAMETER and DATE according to your usage:

Autonomous sea ice measurements (PARAMETER) from DATE to DATE were obtained from https://www.meereisportal.de (grant: REKLIM-2013-04).

3. Buoy data is archived at PANGAEA, a Data Publisher for Earth & Environmental Science, after the buoy has stopped sending data. Please refer to the buoys DOI.